- **Artificial Intelligence (AI)**
  - **Definition:** The broad field of computer science focused on creating systems capable of performing tasks that typically require human intelligence, such as reasoning, decision-making, and pattern recognition (Russell & Norvig, *Artificial Intelligence: A Modern Approach*).
  - **Real-World Example:** Route navigation apps (e.g., Google Maps) using search algorithms like A* to compute optimal driving paths.
- **Machine Learning (ML)**
  - **Definition:** A subfield of AI centered on algorithms that automatically improve performance at a specific task through experience and data, rather than explicit step-by-step programming (Mitchell, *Machine Learning*).
  - **Real-World Example:** Email spam filters that classify incoming messages based on historical patterns in training datasets.
- **Deep Learning (DL)**
  - **Definition:** A specialized subfield of Machine Learning that uses multi-layered artificial neural networks to automatically extract hierarchical features from unstructured raw data (Goodfellow et al., *Deep Learning*).
  - **Real-World Example:** Computer vision models (e.g., Convolutional Neural Networks) used to identify tumors or abnormalities in medical X-rays.

Key Differences

- **Scope and Relationship**
  - **AI:** The overarching field encompassing all intelligent computing.
  - **ML:** A subset of AI focused on learning patterns from data.
  - **DL:** A subset of ML utilizing deep multi-layer neural networks.
- **Feature Engineering**
  - **AI:** Relies on predefined, manually programmed rules or logic.
  - **ML:** Requires human experts to identify and extract relevant features from raw data.
  - **DL:** Automatically extracts and learns complex feature hierarchies directly from input data.

- **Data and Computational Requirements**
    - **AI:** Can run on symbolic logic or traditional heuristics with minimal data.
    - **ML:** Performs effectively on small-to-medium structured datasets using standard hardware.
    - **DL:** Requires large volumes of data and specialized computing power (such as GPUs) to train effectively.
- **Underlying Architecture**
    - **AI:** Knowledge-based systems, expert rules, and decision graphs.
    - **ML:** Statistical models like Decision Trees, Support Vector Machines, and Linear Regression.
    - **DL:** Multi-layered Artificial Neural Networks (>3 layers).